mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA with Circular Positional Encoding

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Abstract

Mitochondrial DNA (mtDNA) is a 16,569 bp circular genome with central roles in human disease, population genetics, and evolutionary biology. Existing DNA foundation models are pre-trained on nuclear genomes and use linear positional encodings, which misrepresent the circular topology of mtDNA: positions 0 and 16,568 are physically adjacent at the D-loop junction but receive maximally different encodings.

We present **mtDNA-FM**, the first foundation model dedicated to mitochondrial DNA. It introduces two domain-specific design choices: (i) a *circular positional encoding* that wraps the positional angle at the genome length, so positions 0 and 16,568 (adjacent across the D-loop junction) receive similar rather than maximally different encodings (positional-encoding cosine similarity $\approx$0.74, versus $\approx$0 under standard linear encoding); and (ii) a *heteroplasmy projection channel* that accepts a continuous per-position variant allele fraction alongside sequence content. The model is pre-trained on 152,590 vertebrate mtDNA sequences in two phases (117,615 cross-species, then 34,975 human HmtDB).

Zero-shot evaluation of the Phase 1 checkpoint achieves 37.9% on 26-class haplogroup classification (95% CI 34.4–41.2%; random baseline 3.85%; 9.8× lift), below both a supervised 6-mer baseline (78.7%) and DNABERT-2 (66.3%), and AUROC 0.777 on pathogenic variant discrimination without labels — a figure that a gene-region-only baseline matches (AUROC 0.75), indicating the signal is largely regional rather than variant-level. Comparison with DNABERT-2 (66.3%, 117M parameters) reveals a clade-specific gap in West Eurasian haplogroups. Per-class analysis yields four observations that identify specific representation failures and map concrete improvements for a next-generation model.

Availability: `https://github.com/vthawfeek/mtdna-foundation-model`, `https://huggingface.co/vthawfeek/mtdna-foundation-model`, `https://huggingface.co/spaces/vthawfeek/mtdna-fm-demo`

1 Introduction

Mitochondrial DNA (mtDNA) is a 16,569 bp circular, double-stranded genome present in hundreds to thousands of copies per cell [Anderson et al., 1981]. mtDNA is maternally inherited without recombination, making it a key tool for phylogenetics, population genetics, and forensic identification [van Oven and Kayser, 2009]. Clinically, pathogenic mtDNA variants cause severe multi-system disorders including MELAS, LHON, and Leigh syndrome, collectively affecting at least 1 in 5,000 individuals [DiMauro and Schon, 2003]. A further complication is *heteroplasmy*: the co-existence of wild-type and mutant mtDNA molecules in the same cell at varying proportions. Disease penetrance for many pathogenic variants depends critically on the heteroplasmy level [Stewart and Chinnery, 2015], yet predicting this from sequence context remains an open problem.

DNABERT-2 [Zhou et al., 2024], HyenaDNA [Nguyen et al., 2023], the Nucleotide Transformer [Dalla-Torre et al., 2025], and Evo [Nguyen et al., 2024] show that pre-training on DNA sequences yields representations that transfer across genomic tasks. All four are pre-trained on nuclear or multi-kingdom genomes without a dedicated mitochondrial corpus or mtDNA-specific design objectives, and share two design gaps for mtDNA:

1. **Linear positional encoding.** Standard sinusoidal PE [Vaswani et al., 2017] treats positions 0 and 16,568 as maximally distant. In the circular mtDNA molecule, these positions are physically adjacent at the D-loop junction. This creates incorrect representations of any functional element spanning the circular boundary.

2. **No heteroplasmy channel.** Existing sequence models assume one definitive base per position. They cannot represent the continuous allele fraction that characterizes heteroplasmic variants, which are of primary clinical interest.

Nuclear DNA also contains repeat elements, splice sites, CpG islands, and regulatory grammar absent from mtDNA. Representations built on billions of bases of nuclear sequence may not transfer to a 16,569 bp genome with different base composition, gene density, and evolutionary constraint.

We present **mtDNA-FM**, a BERT-style [Devlin et al., 2019] foundation model designed and pre-trained for mitochondrial DNA. Our contributions are:

1. A **circular positional encoding** stored as a non-learnable buffer (Section 3.1). Positions 0 and 16,568 are physically adjacent at the D-loop boundary but receive maximally different (cosine similarity $\approx$0) standard sinusoidal encodings. Our circular encoding wraps the base positional angle at the genome length, reducing this mismatch: the two endpoints move one angular step apart ($\Delta\phi \approx 3.8 \times 10^{-4}$ rad) and their full-vector encodings reach cosine similarity $\approx$0.74.

2. A **heteroplasmy projection channel** that accepts per-position variant allele fractions as a continuous input alongside sequence content, designed to let the model represent heteroplasmic variants (Section 3.1). This channel is an architectural provision only: it is inert in the Phase 1 experiments reported here (heteroplasmy input zero everywhere) and is not empirically evaluated.

3. A **two-phase domain-adaptive curriculum**: Phase 1 on 117,615 cross-species vertebrate mtDNA sequences to learn conserved structural features, followed by Phase 2 on 34,975 human HmtDB sequences with heteroplasmy loss enabled (Section 3.2).

4. **Zero-shot evaluation** on 26-class haplogroup classification and pathogenic variant discrimination, with per-class analysis against DNABERT-2 (Section 4).

All evaluations in this preprint use the Phase 1 checkpoint. Ablations of the circular PE, heteroplasmy channel, and Phase 1 vs. Phase 2 curriculum are deferred to the extended paper.

2 Related Work

2.1 Genomic Foundation Models

DNABERT [Ji et al., 2021] was the first BERT-based model for DNA, pre-trained on the human reference genome with 6-mer tokenization. DNABERT-2 [Zhou et al., 2024] extended this to genomes from 32 multi-kingdom species with BPE tokenization and achieved state-of-the-art on the Genome Understanding Evaluation (GUE) benchmark. The Nucleotide Transformer [Dalla-Torre et al., 2025] scales to 2.5B parameters across 850 genomes. HyenaDNA [Nguyen et al., 2023] uses the Hyena implicit convolution operator for million-base context windows. Evo [Nguyen et al., 2024] scales to 7B parameters on prokaryotic and eukaryotic sequences and supports sequence generation and editing. All these models use linear positional encoding, have no heteroplasmy modeling, and were not pre-trained with a dedicated mitochondrial corpus or designed for circular mtDNA topology.

2.2 mtDNA Bioinformatics Tools

Classical mtDNA analysis uses specialized tools: HaploGrep 2 [Weissensteiner et al., 2016] classifies haplogroups from variant calls using PhyloTree [van Oven and Kayser, 2009] branch definitions; Haplocheck [Weissensteiner et al., 2021] detects contamination and heteroplasmy from sequencing data. Pathogenicity databases include MITOMAP [Lott et al., 2013], ClinVar [Landrum et al., 2018], and HelixMTdb [Bolze et al., 2022]. MitImpact [Castellana and Mazza, 2013] aggregates computational pathogenicity scores for protein-coding variants. None of these tools use pre-trained deep learning representations.

2.3 Variant Effect Prediction

Classical methods (SIFT [Ng and Henikoff, 2003], PolyPhen-2 [Adzhubei et al., 2010]) score variants using evolutionary conservation from multiple sequence alignments. Deep generative models (EVE [Frazer et al., 2021], ESM-1v [Meier et al., 2021]) show that pre-trained sequence distributions capture functional constraint. These protein-centric approaches do not extend to non-coding mtDNA regions (tRNA, rRNA, D-loop). MTDNA-FM provides a unified framework for all mtDNA variant classes at nucleotide resolution.

2.4 Positional Encoding Variants

Standard sinusoidal PE [Vaswani et al., 2017] and its learnable counterpart treat position as a linear index. Relative PE methods (RoPE [Su et al., 2024], ALiBi [Press et al., 2022]) encode pairwise distance between tokens, improving length generalization but not circular topology. Sinusoidal and rotary encodings are periodic in phase, but their period is not aligned to any physical sequence length, so a sequence's two endpoints do not become adjacent. To our knowledge, no prior genomic foundation model aligns the positional period to the length of a circular genome so that its two endpoints become adjacent rather than maximally separated; this alignment to circular topology is the specific contribution we claim.

3 Methods

3.1 Architecture

MTDNA-FM is a BERT-style [Devlin et al., 2019] transformer encoder with 6 layers, 8 attention heads, 256-dimensional hidden states, and 1,024-dimensional feed-forward layers (∼5.8M parameters). It uses pre-layer normalisation for training stability [Xiong et al., 2020]. Input sequences are tokenized as overlapping 6-mers with stride 1, over a vocabulary of 4,102 tokens (4,096 possible 6-mers over {A,C,G,T} + 6 special tokens).

3.1.1 Circular Positional Encoding

Standard sinusoidal PE assigns position p an angle proportional to $p/\text{max\_len}$. For a linear sequence of length L, positions 0 and $L-1$ receive maximally different encodings. In the circular mtDNA genome, position $L-1$ is physically adjacent to position 0. We resolve this with a circular angle:

$$\phi_p = \frac{2\pi p}{L_{\text{genome}}} \tag{1}$$

where $L_{\text{genome}} = 16,569$ bp is the rCRS genome length. The positional encoding is then:

$$\text{PE}(p, 2i) = \sin\left(\frac{\phi_p}{10000^{2i/d}}\right) \tag{2}$$

$$\text{PE}(p, 2i+1) = \cos\left(\frac{\phi_p}{10000^{2i/d}}\right) \tag{3}$$

where $d = 256$ is the hidden dimension. The buffer stores positions $p \in \{0, \ldots, L_{\text{genome}} - 1\}$. At the last position $p = L_{\text{genome}} - 1 = 16{,}568$, the base angle is $\phi_{16568} = 2\pi \times 16568/16569 \approx 2\pi - 3.8 \times 10^{-4}$, i.e. within one inter-position step ($\Delta\phi = 2\pi/16569 \approx 3.8 \times 10^{-4}$ rad) of the angle at position 0. Because the geometric frequency schedule $10000^{2i/d}$ is not composed of integer harmonics of the base frequency, only the lowest-frequency channel ($i = 0$) wraps exactly; the higher-frequency channels do not. Across the full 256-dimensional encoding, positions 0 and 16,568 attain cosine similarity ≈0.74, versus ≈0 under standard linear encoding (Figure 2) — roughly a fourfold reduction in the junction discontinuity, though not its elimination. An encoding in which *every* channel wraps would require an integer-harmonic (Fourier) frequency basis, which we leave to future work. The encoding is stored as a non-learnable `nn.Buffer`.

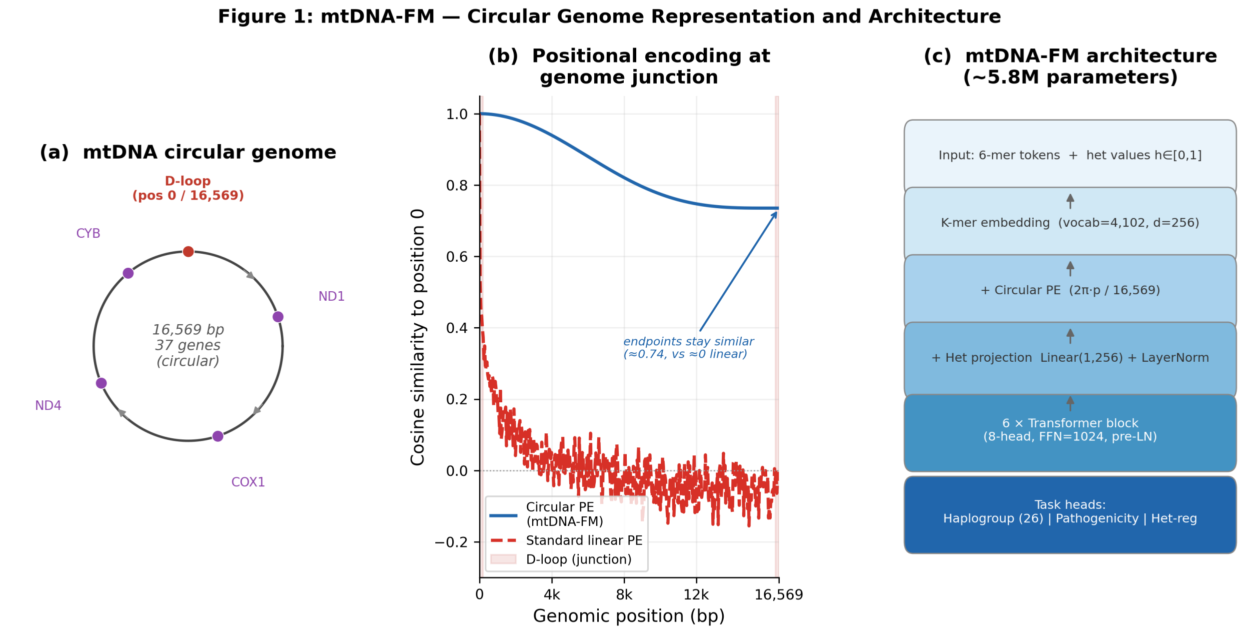

Figure 1: **mtDNA-FM architecture.** (a) The 16,569 bp circular mitochondrial genome with key gene landmarks. (b) Cosine similarity to position 0 across the genome for standard linear PE vs. circular PE. Under the circular encoding the similarity to position 0 stays high near the genome end (endpoint similarity ≈0.74) instead of decaying to ≈0, reducing the artificial positional discontinuity at the D-loop junction. (c) Model block diagram showing the three input channels (k-mer embedding, circular PE, heteroplasmy projection) feeding into 6 transformer layers.

3.1.2 Heteroplasmy Projection Channel

Each input position receives an optional continuous scalar $h_p \in [0, 1]$ representing the variant allele fraction. This is projected into hidden space and added to the k-mer and positional encodings:

$$\mathbf{e}_p = \text{Embed}_{\text{kmer}}(x_p) + \text{PE}(p) + \text{LayerNorm}(\mathbf{W}_{\text{het}}\, h_p + \mathbf{b}_{\text{het}}) \tag{4}$$

where $\mathbf{W}_{\text{het}} \in \mathbb{R}^d$ and $\mathbf{b}_{\text{het}} \in \mathbb{R}^d$ are the learnable weight and bias of an `nn.Linear(1, d)` projection. LayerNorm normalizes the projection output to match the scale of the k-mer embedding. The bias term $\mathbf{b}_{\text{het}}$ is necessary for the channel to encode different heteroplasmy levels: LayerNorm($\mathbf{W}_{\text{het}}\, h_p$) without a bias is invariant to the scale of h_p for any $h_p > 0$, reducing the channel to a binary (present/absent) indicator; the bias $\mathbf{b}_{\text{het}}$ breaks this scale invariance so that distinct values of h_p produce distinct embeddings. When $h_p = 0$ for all positions (Phase 1, where no per-position heteroplasmy targets are available), the channel outputs LayerNorm($\mathbf{b}_{\text{het}}$), a small constant offset relative to the k-mer and positional embeddings.

3.2 Pre-training

3.2.1 Datasets

Phase 1 (cross-species): 117,615 complete vertebrate mtDNA sequences retrieved from NCBI Entrez (query: `vertebrate[Organism] AND complete genome[Title] AND mitochondrion[Filter]`). Sequences span mammals, birds, reptiles, amphibians, and fish.

Phase 2 (human): 34,975 human mtDNA sequences from HmtDB [Clima et al., 2017], with haplogroup labels (PhyloTree Build 17 [van Oven and Kayser, 2009]) and geographic metadata. Sequences not exactly 16,569 bp in length were length-normalized before tokenization: sequences shorter than 16,569 bp had N characters inserted at rCRS position 576 (the 3′ boundary of the control region, immediately upstream of *MT-TF*), preserving downstream protein-coding gene coordinates; sequences longer than 16,569 bp were trimmed

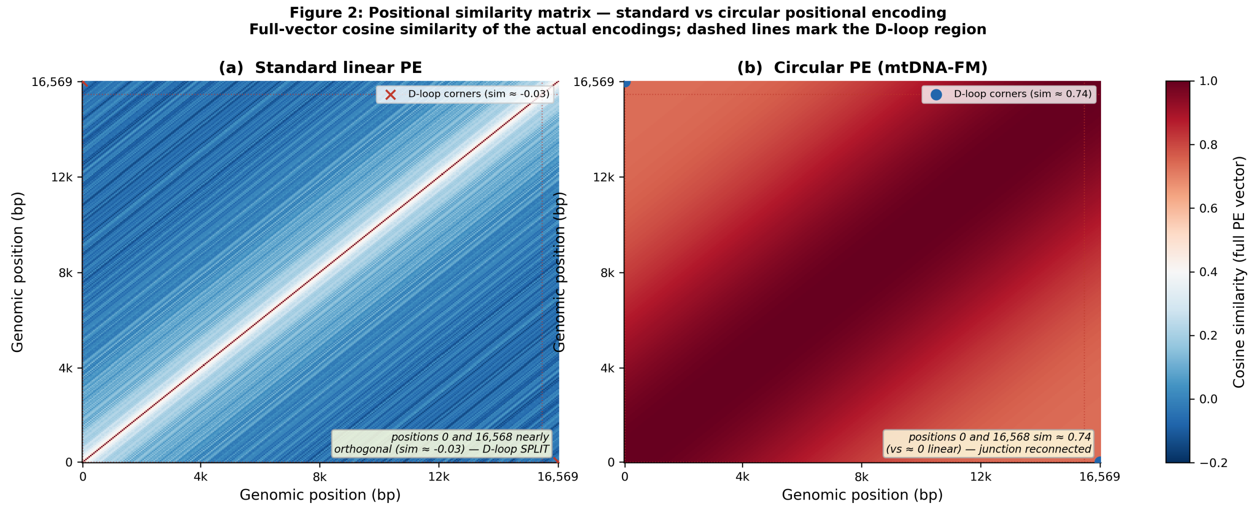

Figure 2: **Positional similarity matrices for standard vs. circular positional encoding.** Each cell (i, j) shows positional similarity between positions i and j. (a) Standard sinusoidal PE: positions 0 and 16,568 are maximally dissimilar, splitting the D-loop across an artificial boundary. (b) Circular PE: positions 0 and 16,568 attain cosine similarity ≈0.74 (vs. ≈0 for linear PE), substantially reducing the junction discontinuity, though the endpoints are not identical. Red dashed lines mark the D-loop region.

from the $3'$ end. A quality filter retaining sequences with at most 10% ambiguous bases after normalization keeps 99.93% of sequences.

3.2.2 Training Objective

Phase 1 uses masked language modeling (MLM; Devlin et al. 2019): 15% of tokens are masked using the 80/10/10 scheme (80% `[MASK]`, 10% random, 10% unchanged). The D-loop homopolymeric C-tract specifically (positions 303–315, 13 positions within the control region, the ∼1,122 bp D-loop spanning positions 16,024–576) is excluded from masking due to high sequencing error rates at homopolymeric runs; the remainder of the D-loop is included in the masked training objective. Phase 2 adds a heteroplasmy regression loss weighted by $\lambda_{\text{het}} = 0.3$:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda_{\text{het}}\, \mathcal{L}_{\text{het}} \tag{5}$$

Heteroplasmy regression targets are derived from the gnomAD v3.1 chrM variant table: for each position with a variant entry, y_{het} is set to the `mean_hl` INFO field (mean variant allele fraction across heteroplasmic carriers at that site), using only variants with at least 50 heteroplasmic carriers (`n_het` ≥ 50). Positions with no gnomAD variant entry or fewer than 50 heteroplasmic carriers receive $y_{\text{het}} = -1$ and are excluded from the het loss. This trains the model to predict population-level heteroplasmy prevalence at known variant positions; per-individual allele-fraction prediction, which requires individual-level sequencing data, is not demonstrated in this preprint.

Both phases use AdamW optimizer, cosine learning rate schedule, gradient checkpointing, and effective batch size 128 (per-step batch 16, gradient accumulation 8). Phase 1: peak LR 1×10^{-4}, 2,000-step warmup, 50,000 steps. Phase 2: peak LR 3×10^{-5}, 500-step warmup, 25,000 steps, with a fresh optimizer state for domain-adaptive pre-training. Phase 1 training converged to a final MLM validation loss of 4.25 (perplexity 70), well below the uniform-prior random baseline of $\ln(4102) = 8.32$ (perplexity 4,102, the vocabulary size), confirming that the encoder learned non-trivial sequence representations prior to any task-specific evaluation.

3.3 Evaluation Setup

Haplogroup zero-shot evaluation. We retrieved 15,741 NCBI human mitochondrial genome records with haplogroup annotations (Entrez query: `human[Organism] AND mitochondrion[Filter] AND complete genome[Title]`

AND haplogroup[Title]), retaining 13,884 sequences with unambiguous major-class assignments across 26 PhyloTree groups. From these we drew a class-balanced subsample (up to 40 sequences per class, to control for the strong natural imbalance) and split it into a train library of 1,509 sequences and a balanced test set of 757 sequences (random_state=42; some rare classes have fewer than 40, minimum 4); the remaining sequences were not used for the zero-shot k-NN. HmtDB was inaccessible at evaluation time, so NCBI sequences were used; these overlap with Phase 1 but not Phase 2 pre-training (see Limitations). No haplogroup labels were used during pre-training.

The zero-shot k-NN uses cosine distance on mean-pooled CLS embeddings across 65 overlapping 512-token windows. All windows use circular indexing: the final window spans positions 16,384–16,568 and 0–326 in a single forward pass, so positions 0 and 16,568 are co-attended at inference, directly exercising the circular PE (Section 3.1.1).

Pathogenicity evaluation. We use 118 ClinVar pathogenic mtDNA SNPs (review status "Pathogenic", "Likely pathogenic", or "Pathogenic/Likely pathogenic"; entries with conflicting interpretations of pathogenicity were excluded) and 419 gnomAD v3.1 chrM common variants (AF > 0.01) as the benign set. Each variant is embedded as the final-layer hidden state of the 6-mer token *starting at* the 0-indexed variant position p (covering positions p to $p+5$ in the variant-mutated rCRS window); with stride-1 tokenization, this token is uniquely identified by position_id $= p$. The input window is constructed from the rCRS reference sequence with the alternate allele substituted at position p; the 512-nucleotide window is centred on p with circular boundary handling (positions beyond 16,569 wrap to the genome start), consistent with the windowing used during pre-training. A 5-fold stratified cross-validated cosine 5-nearest-neighbour classifier is applied: within each fold, the classifier is fit on the 80% training split and the posterior probability (fraction of 5 nearest neighbours that are pathogenic) is recorded for out-of-fold test variants. AUROC and AUPRC are computed from the pooled out-of-fold scores across all 537 variants. Confidence intervals are 1,000-replicate non-parametric bootstrap.

4 Results

4.1 Haplogroup Classification

Zero-shot 5-nearest-neighbour classification on MTDNA-FM embeddings achieves **37.9%** accuracy on 26-class haplogroup prediction (95% CI 34.4–41.2%; random baseline 3.85%; 9.8× lift; macro-F1 0.3703; Table 1). The balanced test set contains 757 sequences across 26 PhyloTree classes drawn from 13,884 NCBI human mitochondrial genomes; the train library contains 1,509 sequences. Note that the evaluation uses a balanced test set (up to 40 sequences per class); accuracy on the natural haplogroup distribution, where haplogroup H alone comprises ∼14% of sequences, has not been evaluated.

Figure 3 shows the UMAP of 100 sequences sampled across 50 sub-haplogroups. African root L clades and the Asian M clade form distinct regions without supervision, whereas the West Eurasian haplogroups (H, HV, J, U, and relatives) overlap heavily (Section 5.3). Misclassifications appear to be phylogenetically structured: H is confused with its ancestor HV; L0 is confused with its sibling L1. The UMAP visualization suggests that cross-clade errors (e.g., African L predicted as West Eurasian H) are rare, as major clades occupy distinct embedding-space regions; formal verification of cross-clade error rates is deferred to the extended paper.

A supervised 6-mer frequency logistic regression on the same 1,509-sequence library achieves 78.7% accuracy (Macro-F1 0.678; Table 1) — higher than either foundation model's frozen-embedding k-NN (DNABERT-2 66.3%, MTDNA-FM 37.9%). For this task, a linear model over raw k-mer counts is therefore a stronger classifier than the pre-trained representations, and MTDNA-FM in particular trails the k-mer baseline by more than twofold. Two conclusions follow. First, the task is easily solvable from surface k-mer statistics, so the low zero-shot accuracy reflects the quality of the learned representation rather than intrinsic task difficulty. Second, and more soberly, the current mtDNA-specialized embeddings do not yet add practical value over a trivial k-mer baseline for haplogroup assignment; closing this gap is the central objective for a scaled successor (Section 5).

Table 1: 26-class haplogroup classification on the same 1,509-sequence train library and 757-sequence balanced test set. Zero-shot methods use frozen embeddings with cosine 5-NN; no haplogroup labels were used in pre-training. MTDNA-FM used 65 overlapping windows (full genome); DNABERT-2 used a single 3,000 nt window (18% of the genome). The 6-mer logistic regression uses the same 1,509-sequence library as the zero-shot methods.

Method	**Setting**	**Accuracy (%)**	**Macro-F1**
Random baseline	26-class	3.85	0.038
MTDNA-FM (zero-shot 5-NN)	26-class	**37.9**	0.3703
DNABERT-2 (zero-shot 5-NN)	26-class	66.3	0.6593
6-mer LR (supervised)	26-class	78.7	0.678

4.2 Variant Pathogenicity Prediction

A 5-fold stratified cross-validated cosine 5-nearest-neighbour classifier on variant-position token embeddings achieves AUROC 0.777 (95% CI 0.731–0.821) against 118 ClinVar pathogenic mtDNA SNPs and 419 gnomAD high-frequency benign variants (AUPR 0.440 vs. 0.220 random; Figure 4). The operating point is TPR 0.69 at FPR 0.21. No pathogenicity labels were used in pre-training.

Per-class AUROC: missense (n_+=56, AUROC 0.726, AUPR 0.303 vs. 0.184 random; 1.6× lift), tRNA (n_+=44, AUROC 0.718, AUPR 0.773), rRNA (n_+=5, AUROC 0.639, limited by sample size). For tRNA variants, the class-specific random AUPR is 44/65 = 0.677 because pathogenic tRNA variants are 68% of the tRNA subset; the observed AUPR 0.773 is only 1.14× above this class-specific baseline, the smallest relative gain among evaluated classes. The overall 2.0× AUPR lift primarily reflects the missense class.

The D-loop class has a single ClinVar pathogenic variant ($n_+ = 1$), which ranked below the median benign variant; AUROC is not reported for a single positive. The "other" variant class (n_+=12) showed below-chance performance (AUROC 0.20); with only 12 positives the bootstrap confidence interval spans nearly the full [0,1] range, so this most likely reflects sampling noise.

The overall metric is confounded by gene region. The pathogenic and benign sets differ sharply in gene-region composition: pathogenic variants are enriched in tRNA genes (37% of pathogenic vs. 5% of benign) and nearly absent from the D-loop (0.8% vs. 26%). Because each variant is embedded at its genomic position in an otherwise-fixed rCRS window, the representation is dominated by position, and position determines region; much of the discrimination is therefore available from region alone. A classifier using *only* the five-way gene-region label (missense, tRNA, rRNA, D-loop, other), with no sequence embedding, achieves AUROC 0.754 and AUPRC 0.463 under the identical 5-fold stratified protocol — statistically indistinguishable from the model's AUROC 0.777 (within its 95% CI) and higher on AUPRC (0.440). At the level of overall ranking, the model does not exceed what gene region alone provides. The scientifically meaningful quantity is *within*-region discrimination, which is modest: AUROC 0.726 within the missense class (AUPR lift 1.6×) and only 1.14× over the base rate within tRNA.

The within-region signal likely reflects evolutionary constraint learned in Phase 1: pathogenic variants concentrate at positions conserved across vertebrates, producing distinct embeddings relative to population-common variants. A proper evaluation must therefore control for region and benchmark against conservation-aware predictors (SIFT, PolyPhen-2, MitoTIP, APOGEE2); this is deferred to the extended paper.

5 Analysis: Four Observations from the Performance Comparison

The per-class results support four observations about how domain-specialized and generalist representations differ. Each identifies a specific cause and rules out simpler explanations. Mechanisms that require ablation to confirm are noted.

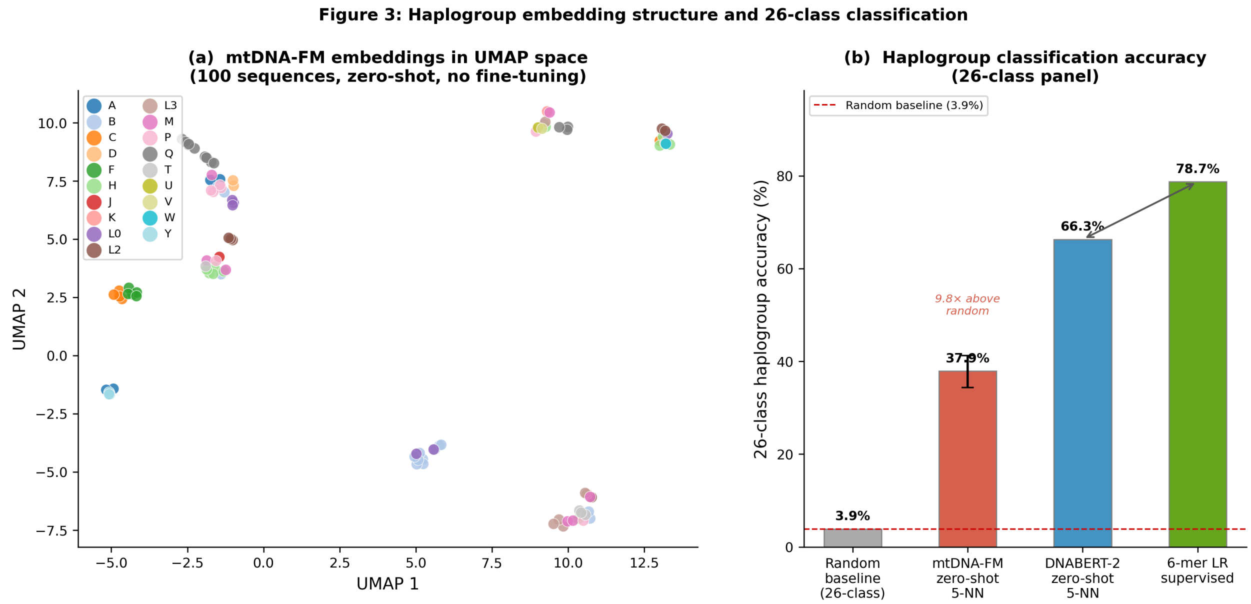

Figure 3: **Haplogroup embedding structure and 26-class classification.** (a) UMAP of mtDNA-FM embeddings for 100 sequences across 50 *sub*-haplogroups (used for visualization only; the classification task uses 26 major PhyloTree classes), coloured by major haplogroup. African L clades and the Asian M clade form separate regions without supervision; the West Eurasian haplogroups (H, HV, J, U, and relatives) overlap heavily, consistent with their near-zero classification F1 (Section 5.3). (b) 26-class haplogroup accuracy across four methods on the same 1,509/757 train/test split. Random baseline 3.85%; mtDNA-FM zero-shot 37.9% (9.8× above random; 95% CI shown); DNABERT-2 zero-shot 66.3%; supervised 6-mer logistic regression 78.7%. Neither mtDNA-FM nor DNABERT-2 used haplogroup labels.

5.1 Observation 1: The Performance Gap Is Clade-Specific

Within mtDNA-FM, all nine West Eurasian haplogroups (H, HV, J, K, T, U, V, W, X) fall in F1 range 0.000–0.241. All six African L haplogroups (L0–L5) fall in 0.438–0.857 (see Table footnote for L5 caveat). These ranges do not overlap: the best West Eurasian haplogroup (U, F1 = 0.241) is strictly below the worst African L haplogroup (L3, F1 = 0.438). Across 15 independent class measurements, complete non-overlap establishes a structural failure specific to West Eurasian haplogroups in mtDNA-FM.

DNABERT-2 shows no analogous separation. Its West Eurasian range (0.400–0.957) and African L range (0.513–1.000) overlap substantially (Table 2).

If the 28.4 percentage-point overall gap were driven purely by scale [Kaplan et al., 2020], performance would improve roughly uniformly across all haplogroups. Instead, the ratio of African L to West Eurasian average F1 is 7.2× within mtDNA-FM and only 1.13× within DNABERT-2. This difference in clade ratios rules out a uniform scale effect.

Table 2: Clade-level average F1 from zero-shot cosine 5-NN (same train/test split). West Eurasian haplogroups: H, HV, J, K, T, U, V, W, X. African L clade: L0–L5. Full per-class values in Supplementary Tables S1 and S3.2.

Clade	**Classes**	**Average F1**		**Gap**	**mtDNA-FM F1 range**
		mtDNA-FM	DNABERT-2		
West Eurasian	9	0.096	0.709	0.613	0.000–0.241
African L	6	0.692	0.803	0.111	0.438–0.857[a]

[a] Upper bound (0.857) is haplogroup L5, $N = 4$ test sequences; excluding L5, the African L range is 0.438–0.816. Non-overlap with West Eurasian range (0.000–0.241) holds either way.

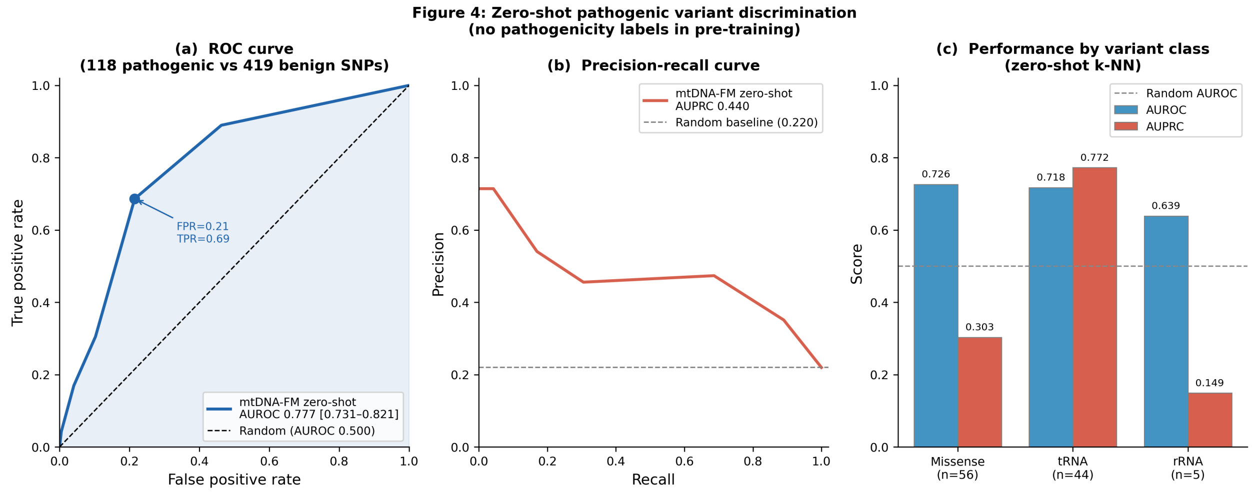

Figure 4: **Pathogenic variant discrimination (no pre-training labels).** (a) ROC curve for 5-fold stratified CV cosine 5-NN on 118 ClinVar pathogenic vs. 419 gnomAD benign mtDNA SNPs. AUROC 0.777 (95% CI 0.731–0.821); operating point at FPR 0.21, TPR 0.69 is marked. (b) Precision-recall curve; AUPRC 0.440 vs. random baseline 0.220. (c) AUROC and AUPRC by variant class: missense (AUROC 0.726), tRNA (AUROC 0.718, AUPRC 0.773; class-specific random AUPR = 0.677), rRNA (AUROC 0.639). No pathogenicity labels were used in pre-training.

5.2 Observation 2: mtDNA-FM's West Eurasian Haplogroup Failure Is a Representation Failure, Not an Information-Access Failure

mtDNA-FM processes all 16,569 positions of each genome across 65 overlapping windows. DNABERT-2 was evaluated on the first 3,000 nt, covering the D-loop control region (positions 1–576), the 12S rRNA gene (648–1601), and the first 1,329 nt of the 16S rRNA gene (1671–3000). The ND1 coding gene does not begin until position 3,307 and falls outside DNABERT-2's window.

Key diagnostic positions for the worst-performing West Eurasian haplogroups fall within the region DNABERT-2 covers. Haplogroup H is defined by positions 2706 and 7028 [van Oven and Kayser, 2009], of which 2706 lies within the first 3,000 nt. Haplogroup J carries a defining control-region variant at position 295 [van Oven and Kayser, 2009], also within this window. Using only these early markers, DNABERT-2 achieves F1 = 0.643 on H and F1 = 0.947 on J.

mtDNA-FM covers the same positions (within the first $\sim$11 of its 65 windows, stride 256, window 512) yet achieves F1 = 0.055 on H and F1 = 0.149 on J. The discriminative information is in positions 1–3,000 nt. DNABERT-2 uses it. mtDNA-FM does not. The failure is therefore in how mtDNA-FM represents what it sees, not in what it has access to.

Three candidate mechanisms are consistent with this observation; each requires ablation to isolate:
(a) Scale. At 5.8M vs 117M parameters, mtDNA-FM has a 20× smaller parameter budget. This reduces representational capacity for subtle sequence differences between closely related haplogroups.
(b) Tokenization. Stride-1 6-mer tokenization creates approximately 16,569 tokens per genome with extreme inter-token autocorrelation: tokens at positions i and $i+1$ share five of six nucleotides. DNABERT-2's BPE vocabulary [Zhou et al., 2024] of approximately 500 non-redundant tokens provides higher information density per token, particularly for haplogroup-diagnostic sequence patterns.
(c) Window mean-pooling. mtDNA-FM uses 65 overlapping 512-token windows with stride 256, producing a single 256-dimensional vector from the mean of 65 window CLS representations. A diagnostic signal at D-loop position 295 contributes equally to a conserved tRNA motif at position 7,450 (*MT-TS1*, tRNA-Ser). This dilutes position-specific markers to roughly 1/65 of the total embedding.

5.3 Observation 3: Bidirectional Precision and Recall Collapse Confirms Embedding Overlap Within the West Eurasian Haplogroup Group

For a k-NN classifier, low recall on class H means H test sequences are assigned to other classes. Low precision means non-H sequences are assigned to H. Both failing simultaneously means H embeddings occupy the same region of cosine space as embeddings from other haplogroups: the k-NN cannot distinguish them.

Table 3 reports precision and recall for West Eurasian haplogroups in MTDNA-FM. Eight of nine show both precision and recall below 0.20 (the sole exception is U, with precision = 0.389). For comparison, haplogroup L0 has precision = 0.857 and recall = 0.750: its embeddings form a distinct cluster. The UMAP in Figure 3 shows this directly: haplogroup H is confused with its ancestor HV, consistent with near-zero F1 for both.

Table 3: Precision and recall for West Eurasian haplogroups under MTDNA-FM zero-shot 5-NN. Low precision and low recall together indicate bidirectional embedding overlap: H sequences are assigned to other classes (low recall) and non-H sequences are assigned to H (low precision). African root clade L0 (precision 0.857, recall 0.750) is shown for comparison.

Haplogroup	Precision	Recall	Support
H	0.061	0.050	40
HV	0.036	0.040	25
J	0.185	0.125	40
K	0.182	0.150	40
T	0.133	0.050	40
U	0.389	0.175	40
V	0.056	0.091	11
W	0.000	0.000	16
X	0.071	0.083	12

5.4 Observation 4: Full-Genome Coverage Provides Measurable Advantage Where DNABERT-2 Is Truncated

Three haplogroups show MTDNA-FM outperforming DNABERT-2: C (0.632 vs 0.611), F (0.716 vs 0.613), and E (0.400 vs 0.222). These haplogroups have PhyloTree-documented diagnostic positions beyond position 3,000 nt [van Oven and Kayser, 2009]. DNABERT-2's single-window evaluation truncates each genome to the first 3,000 nt; MTDNA-FM's 65-window protocol accesses the full genome including these positions.

The result is most reliably supported by haplogroups C ($N = 40$) and F ($N = 36$). Haplogroup E has only 4 test sequences and its result (F1 0.400 vs. 0.222, based on 1–2 additional correct predictions) is anecdotal. This is an evaluation-protocol observation: it is specific to DNABERT-2's single-window evaluation and does not extend to a full-genome DNABERT-2 protocol, which has not been tested.

5.5 Design Implications for Next-Generation mtDNA Models

Each observation identifies a specific improvement target.

Observations 1 and 3 (clade-specific failure and embedding overlap) point to representation quality for recently diverged, sparse-signal clades. Scaling to 100M parameters is the highest-leverage single intervention [Kaplan et al., 2020]. Balanced sampling during Phase 2 pre-training, with haplogroup-uniform weighting, is a lower-cost intervention that directly targets the 14.3% H over-representation in HmtDB.

Observation 2 (representation failure) points to tokenization and aggregation. BPE tokenization trained on human mtDNA sequences would reduce autocorrelated 6-mer tokens to approximately 500 non-redundant tokens [Zhou et al., 2024] and enable a single-window forward pass. Hierarchical attention over window CLS representations would let the model weight windows by diagnostic content, ending the $\sim$1/65 dilution of D-loop signal.

Observation 4 (full-genome advantage under the evaluation protocol used) establishes that full-genome coverage provides measurable advantage when diagnostic positions fall outside a single-window truncation limit. BPE tokenization of the full 16,569 bp genome (approximately 500 tokens) would eliminate windowing entirely, resolving the aggregation problem while keeping full-genome access.

The circular positional encoding addresses the circular topology of mtDNA, a property not handled by any existing genomic model. The heteroplasmy projection channel is an architectural provision for future work on population-level and per-individual allele-fraction modeling. We stress that it is inert in every result reported here: all evaluations use the Phase 1 checkpoint, where the per-position heteroplasmy input is zero everywhere, so the channel reduces to a single constant offset (LayerNorm($\mathbf{b}_{\text{het}}$)) added identically to every token and carries no per-position or per-sequence information (Section 3.1). It should thus be read as an untested design element, not an evaluated contribution. A scaled successor that enables the Phase 2 heteroplasmy objective should preserve both channels and quantify the heteroplasmy channel's effect directly.

6 Discussion

mtDNA-FM shows that a compact 5.8M-parameter encoder pre-trained on mitochondrial sequences encodes some evolutionary structure without supervision, but does not yet yield competitive task performance. Haplogroup identity is recoverable from frozen embeddings at 37.9% 26-class accuracy — above chance, but below both a supervised 6-mer baseline (78.7%) and DNABERT-2 (66.3%). On pathogenic-variant discrimination the model reaches AUROC 0.777, but a gene-region-only baseline matches this (Section 4.2), so the signal is largely regional rather than variant-level. The four observations in Section 5 identify specific causes of these gaps and point to concrete improvements.

Circular PE generalizability. The circular positional encoding is not specific to mtDNA. Any circular DNA molecule benefits from reducing the artificial positional discontinuity at the sequence junction.

Prokaryotic genomes and synthetic biology. Bacterial chromosomes (*E. coli*: 4.6 Mb; *B. subtilis*: 4.2 Mb) and all plasmids are circular. A foundation model with circular PE would correctly represent regulatory elements that span the origin of replication.

Extrachromosomal DNA (ecDNA) in cancer. ecDNA occurs in approximately 14% of human cancers [Turner et al., 2017] as circular elements (0.1–10 Mb) carrying amplified oncogenes and super-enhancers. Its circular topology enables long-range enhancer contacts and rapid copy-number amplification [Kim et al., 2020]. A model with circular PE would correctly represent regulatory grammar at the oncogene–enhancer junction.

Minicircle DNA vectors. Minicircle vectors are small (2–5 kb) circular constructs used in gene therapy [Kay et al., 2010]. The expression cassette forms a closed circle. Circular PE would correctly represent regulatory elements that are physically adjacent across the vector junction.

Ring chromosomes. Ring chromosomes arise from telomeric end fusion into a closed circle. They are associated with clinical syndromes (Ring 14 epilepsy, Ring 20 nocturnal epilepsy, Ring 22 intellectual disability) [Guilherme et al., 2011] and appear in cancer cytogenetics. Circular PE accurately represents the physical topology that linear PE misrepresents at the ring junction.

We provide the circular PE implementation as a reusable `nn.Module` with a documented API.

Why does biologically motivated positional encoding not overcome the performance gap? Three effects explain this. First, the 20× parameter gap (5.8M vs. 117M) dominates: a model without capacity to encode fine-grained haplogroup-discriminating features cannot be rescued by PE alone. Second, the D-loop region (positions 16,024–576, where circular PE matters most) is captured by both models. The critical failure in Observation 2 is in representing diagnostic variants within the first 3,000 nt, not at the junction boundary. Third, under this evaluation protocol (single 3,000 nt window for DNABERT-2; 65-window full-genome for mtDNA-FM), mtDNA-FM outperforms on haplogroups C, F, and E, whose defining markers fall beyond DNABERT-2's truncation limit. A full-genome DNABERT-2 evaluation has not been tested.

Circular PE likely provides a latent advantage at the D-loop junction, currently masked by the parameter budget gap. A controlled ablation comparing circular vs. sinusoidal PE on an otherwise identical model will isolate this contribution in the extended paper. At 5.8M parameters, scale is the dominant variable.

The junction-closure property is also structurally diluted by the evaluation protocol. Of the 65 overlapping 512-token windows that are mean-pooled into each genome embedding, only the one or two that straddle the position-16,568/0 boundary exercise the circular wraparound; within the other ∼63 windows the circular and standard encodings differ only by a global angular compression. The junction benefit therefore reaches at most a few percent of the pooled representation. A single-window model that ingests the full 16,569 bp genome at once (e.g. with BPE tokenization, Section 5) would let the circular encoding act on the junction directly rather than in one or two of 65 averaged windows, and is the setting in which we expect its contribution to be measurable.

Limitations.

- *Phase 1 checkpoint only.* Both evaluations use the Phase 1 checkpoint. Phase 2 training (34,975 HmtDB sequences, heteroplasmy loss) was executed but not evaluated for this preprint due to compute constraints. Phase 1 vs. Phase 2 ablation is deferred to the extended paper.

- *Evaluation leakage and sequence redundancy.* Two distinct overlaps affect the haplogroup evaluation. (i) *Pre-training overlap (* MTDNA-FM *only).* The NCBI human evaluation sequences are a subset of the record type captured by the Phase 1 query (vertebrate complete-genome mtDNA) and thus largely contributed to Phase 1 MLM training; no haplogroup labels were used. The exact fraction cannot be reconstructed — the cross-species corpus was not retained — but it is likely high. This can only advantage MTDNA-FM, which nonetheless trails DNABERT-2 by 28.4 points, so the gap is a lower bound on DNABERT-2's advantage. (ii) *Library–test redundancy (both models).* Human mitochondrial genomes within a haplogroup are near-identical, so the reference library and test queries overlap heavily at the sequence level. Reproducing the exact 1,509/757 split, the median test sequence is 99.93% identical to its nearest library sequence, 4.5% of test sequences have an *exact* library duplicate (differing accession, identical sequence), and 66% have a library twin within ∼16 SNPs. A model-free nearest-sequence lookup (1-NN on raw Hamming distance) consequently classifies haplogroup at 79.5% — more than double MTDNA-FM's 37.9% and above the supervised 6-mer baseline (78.7%). The zero-shot haplogroup task is therefore largely a near-duplicate retrieval problem, on which the model's lossy 256-dimensional embedding underperforms the raw sequence; 37.9% should be read in that light. (Quantified by `paper/experiments/evaluation/eval_overlap_analysis.py`.)

- *European cohort bias.* HmtDB contains approximately 60–70% European-origin sequences; haplogroup H alone is approximately 14.3% of labeled sequences. Whether this over-representation during Phase 2 contributes to embedding-space overlap for West Eurasian haplogroups is a testable hypothesis (Section 5.1).

- *SNPs only.* The pathogenicity and heteroplasmy models handle single-nucleotide variants. Insertions and deletions require a different tokenization strategy. The AF >1% benign proxy may include haplogroup-defining positions that are fixed at high frequency in specific populations; haplogroup-frequency confounding has not been assessed.

- *No wet-lab validation.* All results are computational. Experimental validation of predicted pathogenic variants is future work.

- *Haplogroup resolution.* Sub-haplogroup prediction (H1a vs. H1b) requires finer-grained labels not included in this study.

Future work. The extended paper will include ablations of circular PE vs. sinusoidal PE and the heteroplasmy channel, a Phase 1 vs Phase 2 curriculum ablation, supervised fine-tuning, and a pathogenicity benchmark against MitoTIP and APOGEE2.

7 Conclusion

MTDNA-FM is the first foundation model dedicated to mitochondrial DNA. Its circular positional encoding represents the circular topology of the genome at the D-loop junction more faithfully than standard linear encoding (endpoint cosine similarity ≈0.74 vs. ≈0), a property not addressed by any existing genomic model. Zero-shot evaluation shows the Phase 1 checkpoint encodes some evolutionary structure without supervision (37.9% on 26-class haplogroup classification, 9.8× random), but does not reach competitive task performance: it trails a supervised 6-mer baseline and DNABERT-2 on haplogroups, and its AUROC 0.777 on pathogenic-variant discrimination is matched by a gene-region-only baseline.

Comparison with DNABERT-2 reveals a 28.4 percentage-point gap that is clade-specific and traceable to representation quality, not information access. All nine West Eurasian haplogroups fall in F1 range 0.000–0.241; all six African L haplogroups fall in 0.438–0.857, with no overlap. DNABERT-2 correctly classifies H and J from diagnostic positions that mtDNA-FM also covers, indicating a representation failure rather than a coverage failure. The four observations in Section 5 map four concrete targets for a next-generation model: scaling to 100M parameters, BPE tokenization to reduce token autocorrelation, hierarchical window aggregation, and haplogroup-balanced pre-training. The circular PE and heteroplasmy channel should be preserved; they address structural properties of the genome with no counterpart in existing models.

All code, model weights, and the DVC pipeline are publicly available.

Data Availability

Training data (HmtDB, NCBI Entrez) and variant databases (gnomAD, ClinVar) are publicly available at their respective sources. No proprietary or restricted data were used. Pre-trained model weights are available at `https://huggingface.co/vthawfeek/mtdna-foundation-model`.

Code Availability

All code is available at `https://github.com/vthawfeek/mtdna-foundation-model` under the Apache License 2.0. The full DVC pipeline for end-to-end reproducibility is included.

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Competing Interests

The author declares no competing interests.

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Supplementary Material

See supplementary.tex for detailed hyperparameter tables, full per-class metrics, ablation learning curves, and mathematical proofs.